

SpaceX Falcon 9 First Stage Landing Prediction

IBM Data Science Professional Certificate

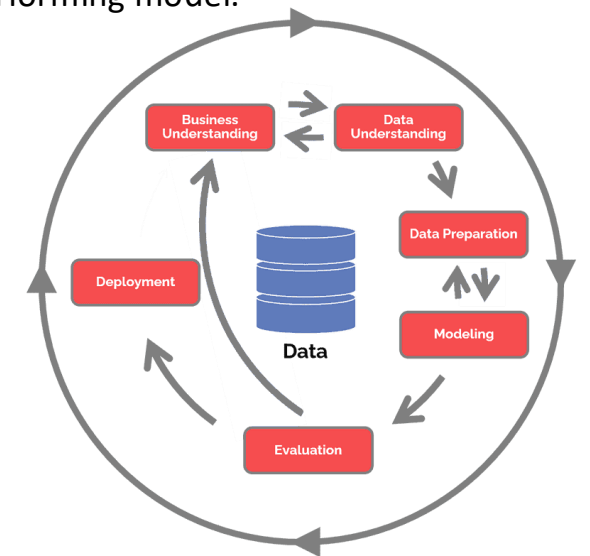
Applied Data Science Capstone

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Executive Summary

- This project focuses on predicting the successful landing of the SpaceX Falcon 9 first stage using data science techniques.
- The project workflow includes data collection, data cleaning, exploratory data analysis, interactive visualization, and machine learning classification.
- Launch data was collected from the SpaceX API and public sources. Python, SQL, Folium, and Plotly Dash were used for analysis and visualization.
- Multiple machine learning models were developed to predict Falcon 9 first-stage landing success.
- The Support Vector Machine (SVM) achieved the highest prediction accuracy of 88%, making it the best-performing model.
- This project demonstrates how data science techniques can effectively predict Falcon 9 first-stage landing success and support data-driven decision making for future launches.



Introduction

- SpaceX Falcon 9 is a reusable rocket system designed to reduce launch costs.
- Successful first-stage landing allows booster reuse and improves mission efficiency.
- Historical launch data can be analyzed to identify factors affecting landing success.

Project Objectives:

- Collect launch data
- Clean and prepare data
- Perform EDA
- Create visualizations
- Build prediction models



Data Collection Methodology

Data Sources:

- SpaceX REST API
- Public web sources

Tools Used:

- Python Requests
- Pandas
- Web Scraping

Collected Features:

- Launch site
- Payload mass
- Orbit type
- Booster version
- Landing outcome

Data Collection Evidence

The collected launch data was converted into a structured dataset using Python and Pandas.

The dataset contains important mission characteristics used for further analysis.

Task 1: Request and parse the SpaceX launch data using the GET request

To make the requested JSON results more consistent, we will use the following static response object for this project:

```
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.js'
```

We should see that the request was successful with the 200 status response code

```
response=requests.get(static_json_url)
```

```
response.status_code
```

```
200
```

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
data = pd.json_normalize(response.json())
```

Using the dataframe `data` print the first 5 rows

```
# Get the head of the dataframe  
data.head()
```

	static_fire_date_utc	static_fire_date_unix	tbd	net	window	rocket	success	details	crew	ships	capsules
0	2006-03-17T00:00:00.000Z	1.142554e+09	False	False	0.0	5e9d0d95eda69955f709d1eb	False	Engine failure at 33 seconds and loss of vehicle	[]	[]	[5eb0e4b5b6c3bb0C
1	NaN	NaN	False	False	0.0	5e9d0d95eda69955f709d1eb	False	Successful first stage burn and transition to second stage, maximum altitude 289 km, Premature engine shutdown at T+7 min 30 s, Failed to reach orbit, Failed to recover first stage	[]	[]	[5eb0e4b5b6c3bb0C

Data Wrangling Methodology

Data preprocessing steps:

- Identified missing values
- Removed unnecessary columns
- Converted data types
- Processed categorical variables
- Prepared dataset for machine learning

Data Cleaning Results

The cleaned dataset improved data quality and ensured reliable analysis and model development.

VIEW LOGS | CODE | MINIMAP

Identify and calculate the percentage of the missing values in each attribute

```
(df.isnull().sum()/len(df))*100
```

[3] ✓ 0.0s Python

```
... FlightNumber    0.000000
Date              0.000000
BoosterVersion    0.000000
PayloadMass       0.000000
Orbit             0.000000
LaunchSite        0.000000
Outcome           0.000000
Flights           0.000000
GridFins          0.000000
Reused            0.000000
Legs              0.000000
LandingPad        28.888889
Block             0.000000
ReusedCount       0.000000
Serial            0.000000
Longitude         0.000000
Latitude          0.000000
dtype: float64
```

handling missing values

handling missing values

```
df["LandingPad"] = df["LandingPad"].fillna("Unknown")
(df.isnull().sum()/len(df))*100
```

[4] ✓ 0.0s Python

```
... FlightNumber    0.0
Date              0.0
BoosterVersion    0.0
PayloadMass       0.0
Orbit             0.0
LaunchSite        0.0
Outcome           0.0
Flights           0.0
GridFins          0.0
Reused            0.0
Legs              0.0
LandingPad        0.0
Block             0.0
ReusedCount       0.0
Serial            0.0
Longitude         0.0
Latitude          0.0
dtype: float64
```

EDA and Data Visualization Methodology

EDA was performed to identify relationships between launch characteristics and landing success.

Tools Used:

- Pandas
- Matplotlib
- Seaborn
- SQL

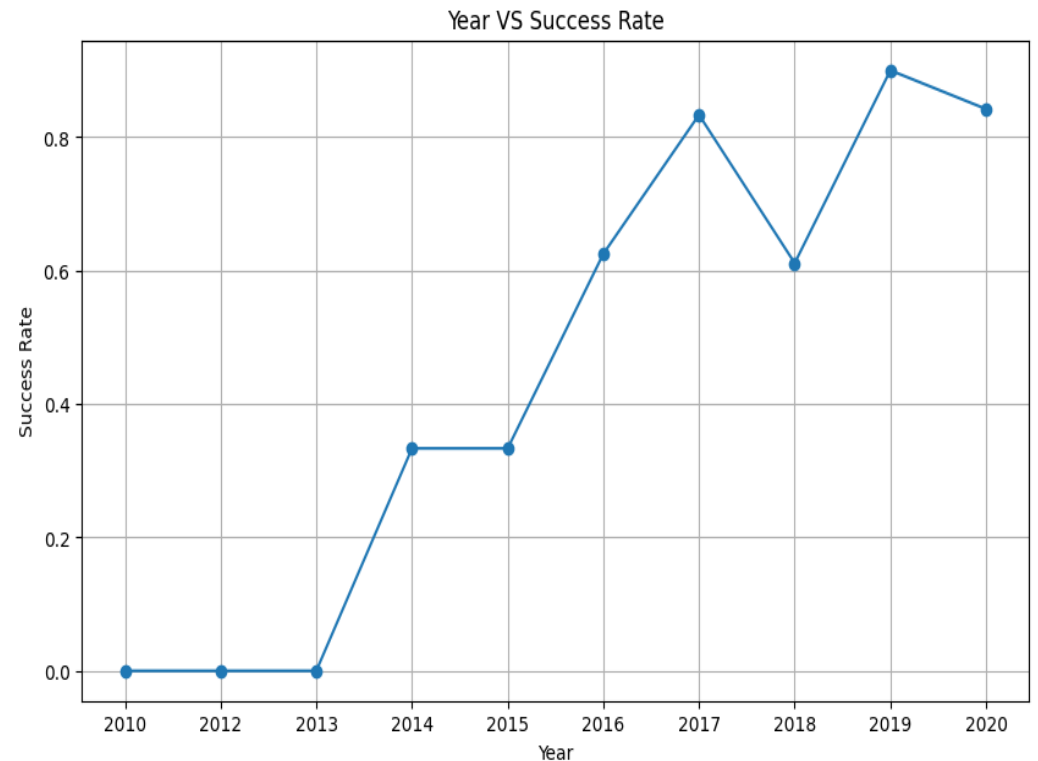
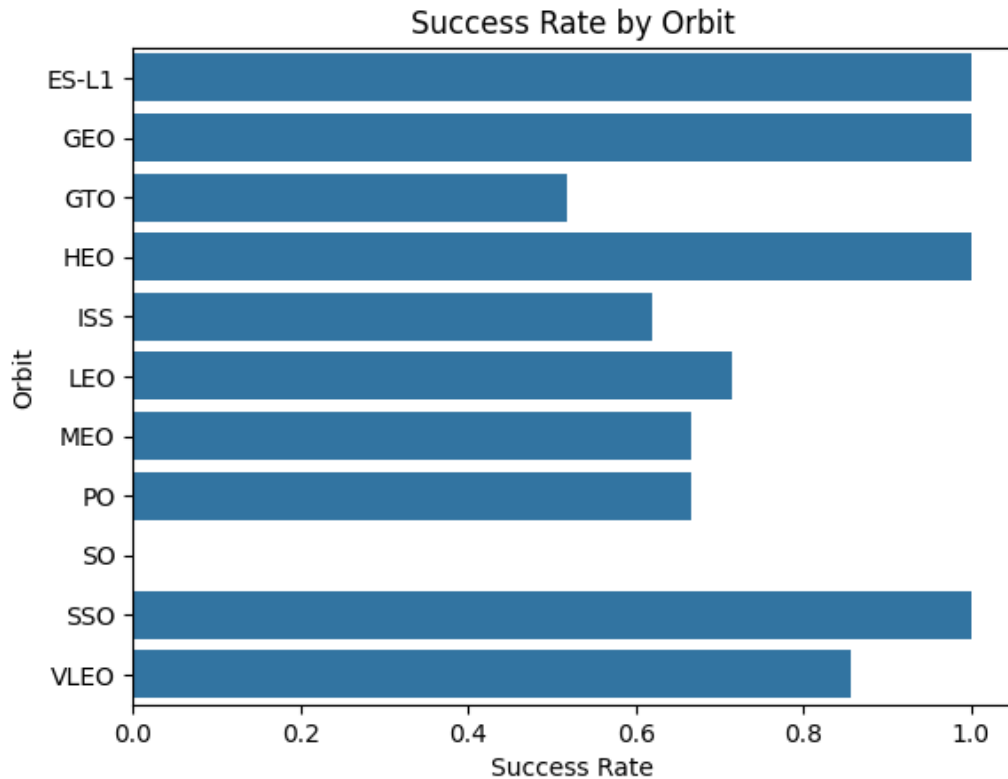
Analysis:

- Launch sites
- Payload mass
- Flight number
- Orbit type

Exploratory Data Analysis Results

Key observations:

- KSC LC-39A recorded the highest landing success rate.
- Landing success increased as booster flight experience increased.
- Payload mass affected the probability of a successful landing



EDA Using SQL Queries

- Methodology
 - SQL queries were used to retrieve, filter, and summarize SpaceX launch records from the database.
- Analysis
 - The queries were performed to analyze mission outcomes, launch sites, payload distribution, and booster landing success.

Key Findings

- SQL efficiently summarized historical launch data.
- The analysis identified launch sites with higher success rates.
- Query results supported the exploratory data analysis and visualization process.

SQL Query Results

SQL analysis helped identify patterns in launch frequency and mission success.

Query 1: Total number of successful and failure missions.

```
pd.read_sql("""
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTBL
GROUP BY Mission_Outcome;
""", conn)
```

Result: Mission outcomes were ranked by total count.

	Mission_Outcome	Total
0	Failure (in flight)	1
1	Success	98
2	Success	1
3	Success (payload status unclear)	1

Launch Site Analysis Using Folium

Methodology

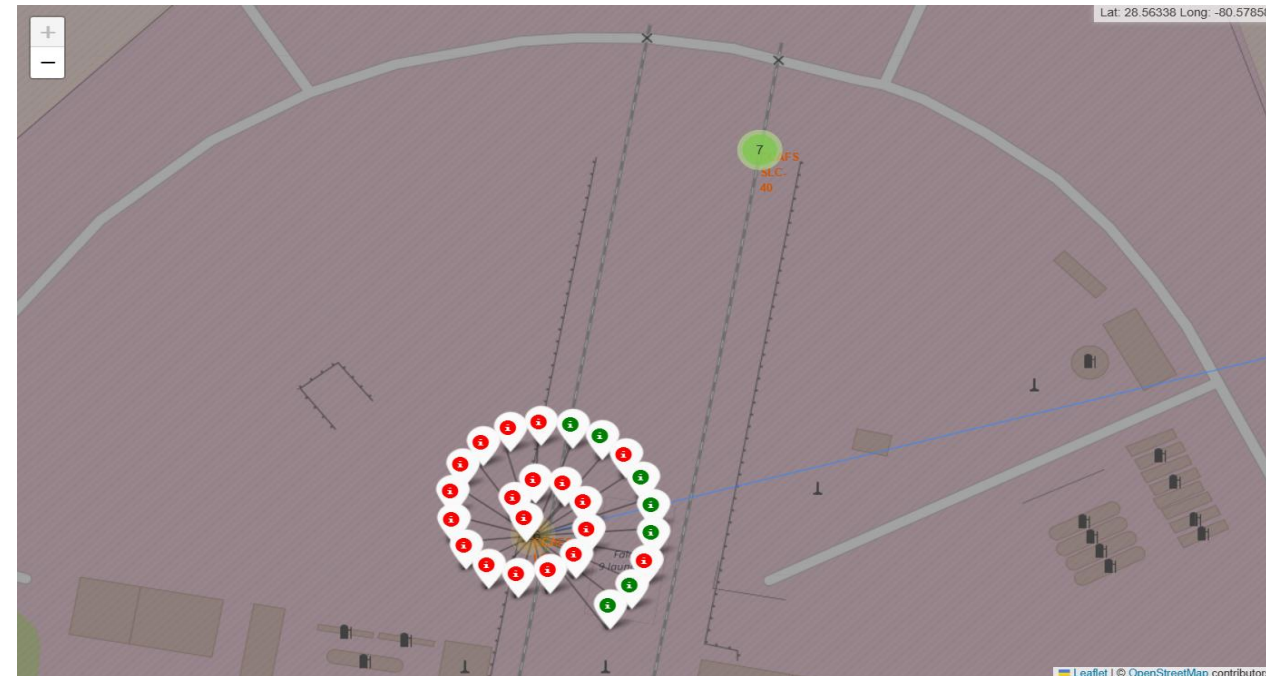
- Folium was used to create an interactive map of SpaceX launch sites using latitude and longitude coordinates obtained during data collection.

Analysis

- The interactive map was used to compare launch site locations and visualize the distribution of successful and unsuccessful booster landings across different SpaceX launch facilities.

Key Findings

- Launch sites are concentrated along the U.S. east coast.
- KSC LC-39A and CCAFS SLC-40 recorded higher successful landing frequencies.
- Launch sites are concentrated along the U.S. east coast.
- KSC LC-39A and CCAFS SLC-40 showed higher landing success rates.



Interactive Dashboard Using Plotly Dash

Methodology

A Plotly Dash dashboard was developed to provide interactive visualization of SpaceX launch data using launch site and payload information.

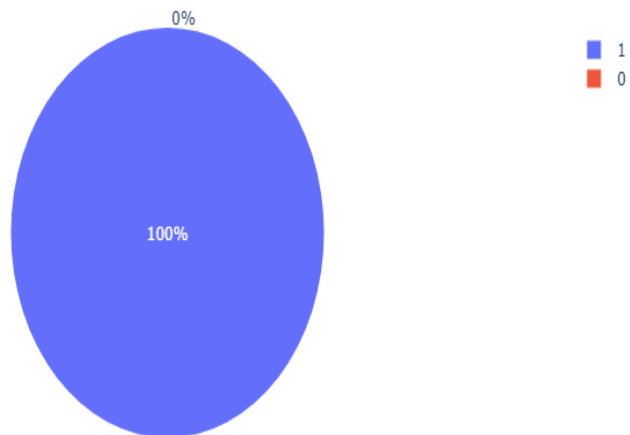
Analysis

The dashboard allows users to filter launches by launch site and payload mass, enabling detailed exploration of the relationship between payload characteristics and landing success.

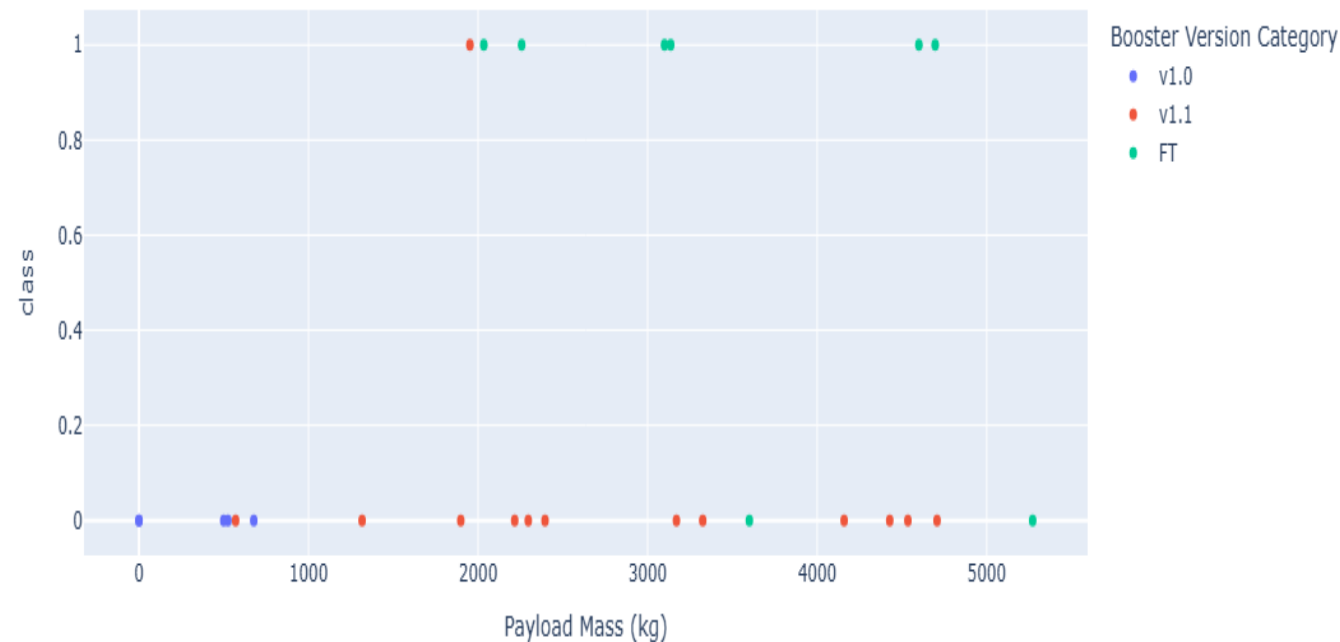
Key Findings

- Landing success varies across launch sites.
- Payload mass influences landing success.
- Interactive filtering helps identify launch performance trends.

Success Rate for CCAFS LC-40



Payload Mass vs Launch Outcome



Predictive Analysis (Classification)

- Methodology
 - Four machine learning algorithms—Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)—were trained and evaluated using historical SpaceX launch data.
- Analysis
 - The performance of each model was compared using prediction accuracy to determine the most effective classifier for predicting first-stage landing success.

Key Findings

- SVM achieved the highest prediction accuracy of 88%.
- Machine learning models successfully predicted Falcon 9 landing outcomes.
- SVM was selected as the final predictive model because it outperformed the other classifiers.

Predictive Analysis Results

Model performance was compared using accuracy scores.
The best-performing model was selected based on prediction accuracy.

Model	Accuracy
Logistic Regression	0.83
Support Vector Machine (SVM)	0.88
Decision Tree	0.78
K-Nearest Neighbors (KNN)	0.83

SVM achieved the highest accuracy (88%) and was selected as the final model because it outperformed the other classifiers.

Conclusion

- SpaceX launch data was successfully collected, cleaned, and analyzed using Python and SQL.
- Exploratory Data Analysis identified important relationships between launch sites, payload mass, and landing success.
- Folium and Plotly Dash provided interactive visualizations that improved understanding of launch performance.
- Machine learning models successfully predicted first-stage landing success.
- Support Vector Machine (SVM) achieved the highest accuracy of 88%, demonstrating the effectiveness of predictive analytics for aerospace applications.

GitHub Repository

Repository:

<https://github.com/kulbhushanranaware/IBM-Data-Science-Capstone>

Repository Includes

- All completed Jupyter notebooks (.ipynb)
- Data Collection notebooks
- Data Wrangling notebook
- SQL notebook
- EDA Visualization notebook
- Folium notebook
- Plotly Dash notebook
- Machine Learning notebook
- README.md
- Final Presentation (PDF)

Thank You

Questions?